Chapter 3.

Sequences – recursion.

Exercise 3A

Write down the next two entries for each of the following.

1.	1,	3,	5,	7,	9,	11,	____,	____.
2.	5,	10,	20,	40,	80,	160,	____,	____.
3.	2,	4,	6,	8,	10,	12,	____,	____.
4.	A,	B,	C,	D,	E,	F,	____,	____.
5.	A,	D,	G,	J,	M,	P,	____,	____.
6.	I,	II,	III,	IV,	V,	VI,	____,	____.
7.	4,	7,	10,	13,	16,	19,	____,	____.
8.	J,	F,	M,	A,	M,	J,	____,	____.
9.	1,	1,	2,	3,	5,	8,	____,	____.
10.	1,	4,	9,	16,	25,	36,	____,	____.
11.	2,	3,	5,	7,	11,	13,	____,	____.
12.	1,	3,	6,	10,	15,	21,	____,	____.
13.	10,	1,	0·1,	0·01,	0·001,	0·0001,	____,	____.
14.	1,	2,	4,	8,	16,	32,	____,	____.
15.	1,	3,	9,	27,	81,	243,	____,	____.
16.	100,	99,	97,	94,	90,	85,	____,	____.
17.	1234,	2341,	3412,	4123,	1234,	2341,	____,	____.
18.	7,	18,	29,	40,	51,	62,	____,	____.
19.	1,	0·5,	0·25,	0·125,	0·0625,	0·03125,	____,	____.
20.	1,	0·5,	$0\cdot\overline{3}$,	0·25,	0·2,	$0\cdot1\overline{6}$,	____,	____.
21.	1,	3,	7,	15,	31,	63,	____,	____.
22.	11,	9,	7,	5,	3,	1,	____,	____.
23.	32,	48,	72,	108,	162,	243,	____,	____.
24.	1,	4,	27,	256,	3125,	46656,	____,	____.

Terms of sequences.

How did you get on with continuing the **sequences** on the previous page?

☞ A **sequence** is a set of items belonging in a certain order, according to some rule. Knowing a particular item and the rule that is involved the next item can be determined.

The previous page involved sequences following many different rules.

Some sequences were probably familiar to you from everyday life:

For example:	A,	B,	C,	D,	E,	F,	...	the letters of the alphabet.
	J,	F,	M,	A,	M,	J,	...	the months of the year.

Others may have been familiar types of number:

For example:	2,	3,	5,	7,	11,	13,	...	the prime numbers.
	1,	4,	9,	16,	25,	36,	...	the square numbers.

Some of the sequences continued by repeatedly adding a number:

For example:	4,	7,	10,	13,	16,	19,	...	adding 3 each time.
	7,	18,	29,	40,	51,	62,	...	adding 11 each time.
	11,	9,	7,	5,	3,	1,	...	adding −2 each time.

Others continued by repeatedly multiplying by a number:

For example:	5,	10,	20,	40,	80,	160,	...	× by 2 each time.
	1,	3,	9,	27,	81,	243,	...	× by 3 each time.
	1,	0·5,	0·25,	0·125,	0·0625,	0·03125,	...	× by 0·5 each time.

Others may have followed other rules.

In this unit we will concentrate on <u>number</u> sequences and we call each number in the sequence a **term** of the sequence.

Thus in the number sequence:

4, 7, 10, 13, 16, 19, ...

the first term is 4, the second term is 7, the third term is 10, the fourth term is 13, etc.

Writing T_1 for the first term, T_2 for the second term and so on we write this as:

$T_1 = 4,\quad T_2 = 7,\quad T_3 = 10,\quad T_4 = 13,\quad T_5 = 16,\quad T_6 = 19,\quad \ldots$

Example 1

For the sequence 5, 11, 23, 47, 95, 191, 383, ... determine

(a) T_3 (b) T_5 (c) $T_3 + T_5$ (d) $3T_2$ (e) $2T_3$ (f) T_8

(a) T_3 is the 3rd term in the sequence. Thus $T_3 = 23$.

(b) T_5 is the 5th term in the sequence. Thus $T_5 = 95$.

(c) $T_3 + T_5 = 23 + 95$
$= 118$

(d) $3T_2 = 3(11)$
$= 33$

(e) $2T_3 = 2(23)$
$= 46$

(f) Noticing that each term is obtained by multiplying the previous term by 2 and then adding 1 it follows that

$$\begin{aligned} T_8 &= 2(T_7) + 1 \\ &= 2(383) + 1 \\ &= 767. \end{aligned}$$

Note: Whilst it is usual to use the letter T for the terms of a sequence, other letters are sometimes used as we shall see in the next exercise.

Exercise 3B

For the sequence 5, 8, 11, 14, 17, 20, 23, 26, 29, ... determine

1. T_4	2. T_5	3. $T_4 + T_5$
4. $3T_4$	5. $4T_3$	6. $T_3 + T_5$
7. $3(T_1 + T_2)$	8. $3T_1 + T_2$	9. T_{10}
10. $T_2 - T_1$	11. $T_3 - T_2$	12. $T_4 - T_3$

For the sequence 3, 7, 15, 31, 63, 127, 255, ... determine

13. T_2	14. T_7	15. $T_2 + T_7$
16. $3T_2$	17. $2T_3$	18. $T_4 + 2T_5$
19. $2T_3 - T_2$	20. $2(T_3 - T_2)$	21. T_8
22. $T_2 - T_1$	23. $T_3 - T_2$	24. $T_4 - T_3$

For the sequence 2, 5, 10, 17, 26, 37, 50, ... determine

25. T_5	26. $3T_2$	27. $T_1 + T_2 + T_3$
28. $T_2 + T_3 + T_4$	29. $T_2 - T_1$	30. $T_3 - T_2$
31. $T_4 - T_3$	32. T_8	

The sequence 1, 1, 2, 3, 5, 8, 13, 21, 34, ... is known as the Fibonacci sequence. For this sequence we tend to use F_1 for the first term, F_2 for the second term etc. Determine

33. F_1	34. F_4	35. $F_1 + F_2$
36. $F_2 + F_3$	37. $F_3 + F_4$	38. $F_6 + F_7 + F_8$
39. $F_6 + F_7 - F_8$	40. F_{10}	

The sequence 1, 3, 4, 7, 11, 18, 29, 47, 76, ... is known as the Lucas sequence. For this sequence we tend to use L_1 for the first term, L_2 for the second term etc. Determine

41. L_1	42. L_4	43. $L_1 + L_2$
44. $L_2 + L_3$	45. $L_3 + L_4$	46. $L_6 + L_7 + L_8$
47. $L_6 + L_7 - L_8$	48. L_{10}	

Recursive rules.

Consider the sequence: 5, 8, 11, 14, 17, 20, 23, ...

Each term is obtained by adding 3 to the previous term.

The statement *Each term is obtained by adding 3 to the previous term* tells us how the sequence **recurs**. It is the **recursive rule** for the sequence.

Knowing the recursive rule for a sequence and at least one term, often the first, allows the terms of the sequence to be determined.

Example 2

Generate the first six terms of a sequence for which:

$$\begin{cases}\text{The first term of the sequence is 73.}\\ \text{Each term is 5 less than the previous term.}\end{cases}$$

The first six terms will be:

	73	$\leftarrow$	1st	term
73 − 5 =	68	$\leftarrow$	2nd	term
68 − 5 =	63	$\leftarrow$	3rd	term
63 − 5 =	58	$\leftarrow$	4th	term
58 − 5 =	53	$\leftarrow$	5th	term
53 − 5 =	48	$\leftarrow$	6th	term

In the above example:

$T_1 = 73$, $T_2 = 68$, $T_3 = 63$, $T_4 = 58$, $T_5 = 53$, $T_6 = 48$.

Writing T_n for the n^{th} term it follows that the next term, T_{n+1}, will be $T_n - 5$.

We can then write the recursive rule *each term is 5 less than the previous term* as

$$T_{n+1} = T_n - 5.$$

To be able to determine the terms of the sequence we also need to know one term. Hence the sequence 73, 68, 63, 58, 53, 48, ... can be defined by stating:

$$T_{n+1} = T_n - 5, \quad T_1 = 73.$$

Read the following and check that you agree with the word description of each recursive formula.

Recursive Formula	Word Description
$T_{n+1} = T_n + 4$	Each term is obtained by adding 4 to the previous term.
$T_{n+1} = T_n - 2$	Each term is obtained by subtracting 2 from the previous term.
$T_{n+1} = 5T_n$	Each term is obtained by multiplying the previous term by 5.
$T_{n+1} = 3T_n + 2$	Each term is obtained by multiplying the previous term by 3 and then adding 2.
$T_{n+2} = T_n + T_{n+1}$	Each term is obtained by adding the two previous terms.

Note
- If the recursive rule $T_{n+1} = T_n - 5$ is written as $T_{n+1} - T_n = -5$ it is called the **difference rule** for the sequence. In this form the rule makes clear the **difference** between any term and the previous one.
- The recursive rule $T_{n+1} = T_n - 5$ could equally well be written in the form

  $$T_n = T_{n-1} - 5.$$

  Both expressions tell us the same thing, i.e. that each term of the sequence is obtained by subtracting 5 from the previous term. (See example 3.)

Example 3

A sequence is defined by $T_n = T_{n-1} + 3$ with $T_1 = 5$.
Determine the first five terms of this sequence.

The formula $T_n = T_{n-1} + 3$ tells us that each term is obtained by adding 3 to the previous term.

Thus if $T_1 = 5$ it follows that

$$\begin{aligned} T_2 &= 5 + 3 = 8, \\ T_3 &= 8 + 3 = 11, \\ T_4 &= 11 + 3 = 14, \\ T_5 &= 14 + 3 = 17. \end{aligned}$$

The first five terms are 5, 8, 11, 14, 17.

Example 4

A sequence is defined by $T_{n+1} = 2T_n + 3$ with $T_1 = 6$.
Determine the first five terms of this sequence.

The formula $T_{n+1} = 2T_n + 3$ tells us that each term is obtained by multiplying the previous term by 2 and then adding 3.

Thus if $T_1 = 6$ it follows that

$$\begin{aligned} T_2 &= 2 \times 6 + 3 = 15, \\ T_3 &= 2 \times 15 + 3 = 33, \\ T_4 &= 2 \times 33 + 3 = 69, \\ T_5 &= 2 \times 69 + 3 = 141. \end{aligned}$$

The first five terms are 6, 15, 33, 69, 141.

Example 5

A sequence is defined by $T_n = 0{\cdot}5T_{n-1}$ with $T_1 = 400$. Determine the first five terms of this sequence.

The formula $T_n = 0{\cdot}5T_{n-1}$ tells us that each term is obtained by multiplying the previous term by 0·5.

Thus if $T_1 = 400$ it follows that

$$\begin{aligned} T_2 &= 200, \\ T_3 &= 100, \\ T_4 &= 50, \\ T_5 &= 25. \end{aligned}$$

The first five terms are 400, 200, 100, 50, 25.

Example 6

A sequence is defined by $T_{n+1} = 4T_n - 2$ with $T_1 = 3$. Determine the first five terms of this sequence.

The formula $T_{n+1} = 4T_n - 2$ tells us that each term is obtained by multiplying the previous term by 4 and then subtracting 2.

Thus if $T_1 = 3$ it follows that

$$\begin{aligned} T_2 &= 4(3) - 2 = 10, \\ T_3 &= 4(10) - 2 = 38, \\ T_4 &= 4(38) - 2 = 150, \\ T_5 &= 4(150) - 2 = 598. \end{aligned}$$

The first five terms are 3, 10, 38, 150, 598.

Alternatively, rather than thinking and interpreting the recursive formula

$$T_{n+1} = 4T_n - 2$$

as meaning *multiply the previous term by 4 and then subtract 2*, we could simply substitute into the given formula:

To obtain T_2 substitute $n = 1$ into	$T_{n+1} = 4T_n - 2$	$T_2 = 4T_1 - 2 = 10$
To obtain T_3 substitute $n = 2$ into	$T_{n+1} = 4T_n - 2$	$T_3 = 4T_2 - 2 = 38$
To obtain T_4 substitute $n = 3$ into	$T_{n+1} = 4T_n - 2$	$T_4 = 4T_3 - 2 = 150$
To obtain T_5 substitute $n = 4$ into	$T_{n+1} = 4T_n - 2$	$T_5 = 4T_4 - 2 = 598$

Note • This substitution method may seem rather formal and tedious but it can be useful if the recursive formula is not easy to understand intuitively. (See example 7).

Example 7

A sequence is defined by $T_{n+1} = 2^n T_n$ with $T_1 = 5$. Determine the first four terms of this sequence.

With $n = 1$ $T_{n+1} = 2^n T_n$ becomes

$$\begin{aligned} T_2 &= 2^1 T_1 \\ &= 2 \times 5 \\ &= 10 \end{aligned}$$

With $n = 2$ $T_{n+1} = 2^n T_n$ becomes

$$\begin{aligned} T_3 &= 2^2 T_2 \\ &= 4 \times 10 \\ &= 40 \end{aligned}$$

With $n = 3$ $T_{n+1} = 2^n T_n$ becomes

$$\begin{aligned} T_4 &= 2^3 T_3 \\ &= 8 \times 40 \\ &= 320 \end{aligned}$$

The first four terms are 5, 10, 40, 320.

Alternatively the recursive formula could be put into a calculator and the terms of the sequence displayed, as shown on the right for the previous example.

Note that the display on the right also shows the progressive sums (also called partial sums).

$T_1 = 5$

$T_1 + T_2 = 5 + 10 = 15$

$T_1 + T_2 + T_3 = 5 + 10 + 40 = 55$

etc.

Calculators differ in the way they accept and display such information. If you wish to use a calculator in this way make sure that **you** can use **your** calculator to input recursive formulae and to display the terms of a sequence.

☑ $a_{n+1} = 2^n \cdot a_n$
$a_1 = 5$

☐ $b_{n+1} = \square$
$b_1 = 0$

☐ $c_{n+1} = \square$
$c_1 = 0$

n	a_n	Σa_n
1	5	5
2	10	15
3	40	55
4	320	375
5	5120	5495

Whilst it is usual to give the first term of a sequence with the recursive rule, knowing any term will allow other terms to be determined. For example the statements:

$$\begin{cases} \text{The fourth term of the sequence is 13.} \\ \text{Each term is 3 more than the previous term.} \end{cases}$$

also allows the sequence 4, 7, 10, 13, 16, 19, 22, ... to be determined.

Example 8

Generate the first six terms of a sequence for which:

$$\begin{cases} \text{The fourth term of the sequence is 47.} \\ \text{Each term is twice the previous term add one.} \end{cases}$$

The task is to determine T_1 to T_6 given the following information:

T_1	T_2	T_3	T_4	T_5	T_6
			47		

× 2 + 1 × 2 + 1 × 2 + 1 × 2 + 1 × 2 + 1

Hence
$$\begin{aligned} T_5 &= 2T_4 + 1 \qquad & T_6 &= 2T_5 + 1 \\ &= 95 & &= 191 \end{aligned}$$

The third term, and similarly the second and first, could be found intuitively by thinking what number when doubled and 1 is added gives an answer of 47.

Alternatively an algebraic approach is possible, as follows:

$$\begin{array}{llll} & T_4 = 2T_3 + 1 & T_3 = 2T_2 + 1 & T_2 = 2T_1 + 1 \\ & 47 = 2T_3 + 1 & 23 = 2T_2 + 1 & 11 = 2T_1 + 1 \\ \therefore & 46 = 2T_3 & 22 = 2T_2 & 10 = 2T_1 \\ \text{i.e.} & T_3 = 23 & T_2 = 11 & T_1 = 5 \end{array}$$

The first six terms of the sequence are 5, 11, 23, 47, 95, 191.

Example 9

State T_1 and the recursive formula for the sequence 98, 91, 84, 77, 70, ...

By inspection each term in the sequence is 7 less than the term before it.
Thus $T_{n+1} = T_n - 7$ with $T_1 = 98$.

Example 10

A sequence has a first term of 240 and each term after the first is equal to the previous term increased by 10%.
State T_1 to T_4 and the recursive formula for the sequence.

Each term after the first will be the previous term multiplied by 1·1.
Thus $T_1 = 240$, $T_2 = 264$, $T_3 = 290{\cdot}4$, $T_4 = 319{\cdot}44$ and $T_{n+1} = 1{\cdot}1 \times T_n$.

Example 11

Given that the sequence 3, 17, 73, 297, 1193, ... has a recursive formula that is of the form $T_{n+1} = aT_n + b$ determine a, b and T_6.

The values of a and b can be obtained by thoughtful trial and adjustment:
The given terms are all integers which suggests that a and b are integers.

With $T_1 = 3$ and $T_2 = 17$, try a = 1 and b = 14. Then $T_3 = 1(17) + 14 \neq 73$

Try a = 2 and b = 11. Then $T_3 = 2(17) + 11 \neq 73$

Try a = 3 and b = 8. Then $T_3 = 3(17) + 8 \neq 73$

Try a = 4 and b = 5. Then $T_3 = 4(17) + 5 = 73$ ✓

and $T_4 = 4(73) + 5 = 297$ ✓

Hence $T_6 = 4T_5 + 5 = 4(1193) + 5$
$= 4777$

The required values are a = 4, b = 5 and $T_6 = 4777$.

Alternatively, using an algebraic approach:
From $T_{n+1} = aT_n + b$ we know that each term is obtained by multiplying the previous term by a and then adding b.

Thus with $T_1 = 3$ and $T_2 = 17$ it follows that $17 = 3a + b$, ←[1]
and with $T_2 = 17$ and $T_3 = 73$ it follows that $73 = 17a + b$. ←[2]

Solving equations [1] and [2] simultaneously gives a = 4 and b = 5 as before.

Exercise 3C

1. For each description, find an appropriate recursive formula from the list.

<u>Descriptions</u>

a : Each term is obtained by adding 5 to the previous term.
b : Each term is obtained by subtracting 5 from the previous term.
c : Each term is obtained by multiplying the previous term by 5.
d : Each term is obtained by multiplying the previous term by 0·5.
e : Each term is obtained by adding the two previous terms.
f : Each term is obtained by multiplying the previous term by 2 and then subtracting 5.
g : Each term is obtained by adding the three previous terms.
h : Each term is obtained by dividing the previous term by 2.
i : Each term is the product of the previous three terms.

<u>Recursive Formulae</u>

A:	$T_{n+1} = T_n - 5$	B:	$T_{n+1} = 5T_n$
C:	$T_{n+1} = (T_n)^5$	D:	$T_{n+1} = 0{\cdot}5(T_n)$
E:	$T_{n+2} = T_n - T_{n+1}$	F:	$T_{n+3} = (T_n)(T_{n+1})(T_{n+2})$
G:	$T_{n+1} = T_n + T_{n-1}$	H:	$T_{n+3} = T_n + T_{n+1} + T_{n+2}$
I:	$T_{n+1} = T_n + 5$	J:	$T_{n+1} = 5T_n + 2$
K:	$T_{n+1} = 2T_n - 5$	L:	$T_{n+1} = 5T_n - 2$

2. Determine the first six terms of a sequence for which:

   The first term of the sequence is 11.
   Each term is five more than the previous term.

3. Determine the first six terms of a sequence for which:

   The first term of the sequence is 4.
   Each term is the previous term doubled and then take away 5.

4. A sequence is defined by $T_n = T_{n-1} + 4$ with $T_1 = 3$. Determine the first five terms of this sequence.

5. A sequence is defined by $T_{n+1} = T_n + 3$ with $T_1 = 1$. Determine the first five terms of this sequence.

6. A sequence is defined by $T_{n+1} = 2T_n + 1$ with $T_1 = 2$. Determine the first five terms of this sequence.

7. A sequence is defined by $T_{n+1} = 3T_n - 1$ with $T_1 = 2$. Determine the first five terms of this sequence.

8. A sequence is defined by $T_n = 5T_{n-1} - 2$ with $T_1 = 1$. Determine the first five terms of this sequence.

9. A sequence has a first term of 128 and each term after the first is equal to the previous term increased by 25%.
State T_1 to T_4 and the recursive formula for the sequence.

10. A sequence has a first term of 1000 and each term after the first is equal to the previous term increased by 20%.
State T_1 to T_4 and the recursive formula for the sequence.

11. A sequence has a first term of 2000 and each term after the first is equal to the previous term decreased by 15%.
State T_1 to T_4 and the recursive formula for the sequence.

12. State T_1 and a recursive formula for the sequence 5, 9, 13, 17, 21, ...

13. State T_1 and a recursive formula for the sequence 26, 23, 20, 17, 14, ...

14. State T_1 and a recursive formula for the sequence 6, 13, 27, 55, 111, ...

15. A sequence has a recursive formula of the form $T_n = 2T_{n-1} + a$, with $T_1 = 3$.
If $T_2 = 10$ find the value of a and determine T_3.

16. Determine the first six terms of a sequence for which:
$\begin{cases} \text{The second term of the sequence is 23.} \\ \text{Each term is eight more than the previous term.} \end{cases}$

17. Determine the first six terms of a sequence for which:
$\begin{cases} \text{The third term of the sequence is 74.} \\ \text{Each term is three times the previous term take 4.} \end{cases}$

18. Determine the first six terms of a sequence for which:
$\begin{cases} \text{The fourth term of the sequence is 89.} \\ \text{Each term is twice the previous term take 1.} \end{cases}$

19. A sequence is defined by $T_{n+2} = T_n + T_{n+1}$ with $T_1 = 1$ and $T_2 = 4$. Determine the first five terms of this sequence.

20. A sequence is defined by $T_n = T_{n-2} + T_{n-1}$ with $T_1 = T_2 = 7$. Determine the first five terms of this sequence.

21. A sequence is defined by $T_{n+2} = 2T_n - T_{n+1}$ with $T_1 = 3$ and $T_2 = 4$. Determine the first five terms of this sequence.

22. A sequence is defined by $T_n = T_{n-3} + T_{n-2} + T_{n-1}$ with $T_1 = 1$, $T_2 = 2$ and $T_3 = 3$. Determine the first six terms of this sequence.

23. Given that the sequence 1, 8, 43, 218, 1093, ... has a recursive formula that is of the form $T_{n+1} = aT_n + b$ determine a, b and T_6.

24. Given that the sequence 5, 7, 11, 19, 35, ... has a recursive formula that is of the form $T_{n+1} = aT_n + b$ determine a, b and T_6.

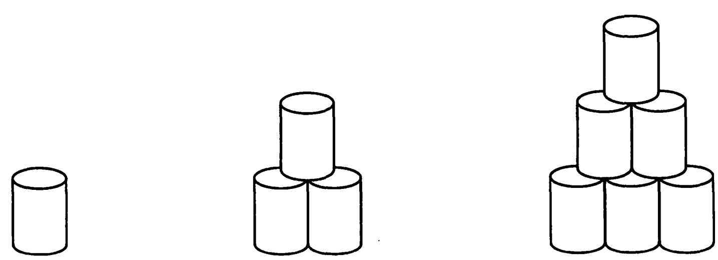
25. Sequence A has terms $A_1, A_2, A_3, \ldots$ and sequence B has terms $B_1, B_2, B_3, \ldots$. Given that $A_1 = 5$, $A_n = A_{n-1} + 3$, $B_1 = 64$, and $B_n = 0{\cdot}75B_{n-1}$, determine the first five terms of sequence C given that $C_n = A_n + B_n$.

26. A sequence has the recursive formula $T_{n+1} = (-1)^n\, 2T_n$, with $T_1 = 1$. Determine the first five terms of this sequence.

27. A sequence has the recursive formula $T_{n+1} = (-1)^{n+1} 2T_n$, with $T_1 = 3$. Determine the first five terms of this sequence.

28. A sequence has the recursive formula $T_{n+1} = (n+2)^n T_n$, with $T_1 = 3$. Determine the first five terms of this sequence.

29. Stacking Cans



(a) Copy and complete the table shown below.

Stacking Cans					
T_1	T_2	T_3	T_4	T_5	T_6
1	3	6	10		

(b) Which of the following recursive formulae give the same first six terms as you obtained in the table above?

I: $T_{n+1} = T_n + 2$, $T_1 = 1$.

II: $T_n = T_{n-1} + 2$, $T_1 = 1$.

III: $T_n = 2T_{n-1}$, $T_1 = 1$.

IV: $T_n = T_{n-1} + n$, $T_1 = 1$.

Spreadsheets.

Computer spreadsheets can be very useful for generating and displaying the terms of a sequence.

Create the following table on a spreadsheet and, by following each pattern down, complete cells A6, A7, A8, B6, B7, B8, C6, C7, C8.

	A	B	C	D
1	1	6	5	
2	2	7	10	
3	3	8	15	
4	4	9	20	
5	5	10	25	
6				
7				
8				

① The entry that is to go in cell D1 is to be given by the rule:

D1 = A1*B1- C1 (where * represents multiply).

Similarly DN = AN*BN- CN

Complete cells D1 to D8 on your spreadsheet.

② The numbers in cells D1 to D8 are the first 8 ______ numbers. (What word goes in the space?)

③ If columns A, B and C were continued indefinitely, and column D was filled according to the rule in part ① above, would column D continue to give the type of numbers mentioned in ② above? Prove it.

Miscellaneous Exercise Three.

This miscellaneous exercise may include questions involving the work of this chapter, the work of any previous chapters, and the ideas mentioned in the preliminary work section at the beginning of the book.

1. Write down the next two entries for each of the following sequences.

(a) 1, 4, 13, 40, 121, 364, ______, ______.

(b) 1, 8, 27, 64, 125, 216, ______, ______.

(c) 1000, 800, 640, 512, 409·6, 327·68, ______, ______.

(d) 1000, 800, 600, 400, 200, 0, ______, ______.

(e) $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{2}$, $\frac{5}{8}$, $\frac{3}{4}$, ______, ______.

2. A sequence is defined by $T_{n+1} = 7T_n - 5$ with $T_1 = 1$.
   Determine the first four terms of this sequence.

3. A sequence is defined by $T_n = 0{\cdot}5T_{n-1} - 2$ with $T_1 = 120$.
   Determine the first four terms of this sequence.

4. At various places in this book we have considered the food consumption / heart disease death rate situation. We found that with y as the deaths due to heart disease, in deaths per 100 000 of population, and x as the foodstuff consumption, in kilograms per person per year, the association could be quite well modelled by the least squares regression line
   The data collected suggested that the situation could be well modelled using a line of best fit with equation
   $$y = 3{\cdot}1x + 90{\cdot}9.$$
   Interpret the numbers 3·1 and 90·9 in this situation.

5. A sequence has a recursive formula of the form $T_n = aT_{n-1} + 3$, with $T_1 = 2$.
   Given that $T_2 = 25$ find the value of a and determine T_3.

6. Sequence D has terms $D_1, D_2, D_3, \ldots$ and sequence E has terms $E_1, E_2, E_3, \ldots$.
   Given that $D_1 = 2$, $D_n = D_{n-1} + 4$, $E_1 = 10$, and $E_n = E_{n-1} - 3$, determine the first five terms of sequence F given that $F_n = (D_n)(E_n)$.

7. A scientific experiment involved a student collecting bivariate data for the two variables x and L. The tabulated results were as follows:

x	0·5	1·0	1·5	2·0	2·5	3·0	3·5	4·0
L	6·9	7·4	9·2	13·4	19·5	28·3	41·3	58·1

   Applying linear regression techniques to the data the student found that the correlation coefficient was 0·93 and the least squares regression line had equation:
   $$L = 14{\cdot}1x - 8{\cdot}7\,.$$
   The student concluded that linear modelling of the situation was the most appropriate and that $L = 14{\cdot}1x - 8{\cdot}7$ was the most appropriate equation to use to represent the relationship algebraically.
   Comment on this conclusion.

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8. The table below gives the *Number of pupils per teacher* (in first year of schooling) and the *Adult Literacy Rate* (percentage of adult population who are literate) for 22 countries.

Country	A	B	C	D	E	F	G	H	I	J	K
Pupils per teacher	22	28	27	41	26	18	25	51	37	25	19
Literacy Rate (%)	95	54	92	51	88	99	65	54	98	60	85

Country	L	M	N	O	P	Q	R	S	T	U	V
Pupils per teacher	36	19	46	16	31	23	63	27	31	67	36
Literacy Rate (%)	67	95	48	99	35	81	35	90	87	30	96

(a) Display the data as a scattergraph on either a calculator display or spreadsheet facility with the pupils per teacher on the horizontal axis and literacy rate on the vertical axis.
(b) On the basis of this data could we conclude that a high literacy rate is due to a low number of pupils per teacher?
(c) Estimate the literacy rate for a country with 55 pupils per teacher.

9. (a) In the section about the line of best fit the equation of the least squares regression line of y on x was expressed as:

$$y - \bar{y} = \frac{s_{xy}}{s_x^{\,2}}(x - \bar{x}) \; .$$

Prove that this line must pass through the point $(\bar{x}, \bar{y})$.

(b) Why is the line of best fit referred to as the **"least squares"** regression line?

10. In 2013, in Western Australia, the numbers of year 12 students of each gender achieving the various grades in the top unit of Mathematics (non-specialist) and the top unit in Biology, were as follows:

Mathematics:

	A	B	C	D	E	Total
Male	616	558	867	253	68	2362
Female	415	383	569	140	25	1532

School Curriculum and Standards Authority. 2013 Year 12 Student Achievement Data. Reproduced with permission from the Government of Western Australia, School Curriculum and Standards Authority.

Biology:

	A	B	C	D	E	Total
Male	77	144	306	72	7	606
Female	214	313	415	71	14	1027

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Analyse, comment, report.